Unit 2: Electromagnetic Induction

"Nature Abhors a Change in Flux"

Part 1: The Basics (Magnetism Reset)

Since we missed this last year, here is the absolute minimum you need to know to survive Induction.

1.1 The Magnetic Field ($\vec{B}$)

Just as mass creates a gravity field, moving charges (currents) create a magnetic field.

- **Symbol:** $\vec{B}$
- **Unit:** Tesla (T)
- **Visual:** Field lines exit **North**, enter **South**.
    - *Density of lines = Strength of Field.*

1.2 Creating Fields (The Sources)

You will see three main sources in problems. Know these formulas (from Ampere's Law):

Source	Geometry	Formula for B	Uniform?
Straight Wire	Concentric circles	$B = \frac{\mu_0 I}{2\pi r}$	No (weaker with distance)
Solenoid (Coil)	Parallel lines inside	$B = \mu_0 \frac{N \cdot I}{L}$	**YES** (Inside the core)
Flat Coil (Loop)	Donuts around wire	$B = \frac{\mu_0 NI}{2R}$	No (Only at center)

> **Constant:** Permeability of free space $\mu_0 = 4\pi \times 10^{-7}\, T \cdot m/A$

Part 2: Magnetic Flux (Φ)

The "Raincatcher" Concept

Before we can have induction, we must measure "how much" magnetism passes through our circuit. This is **Flux**.

2.1 The Definition

$$\Phi = \vec{B} \cdot \vec{S} = NBS\cos(\theta)$$

- **Φ (Phi):** Magnetic Flux [Weber, Wb]
- **N:** Number of turns in the coil.
- **B:** Magnetic Field Strength [Tesla, T]
- **S:** Area of the loop [m^2]
- **θ:** The angle between the Field $\vec{B}$ and the **NORMAL VECTOR** $\vec{n}$.

⚠️ The Danger Zone: Measuring θ

The most common mistake in physics exams involves the angle.

- **$\vec{n}$ (Normal):** A vector strictly perpendicular (90°) to the surface of the coil.
- **If the problem says:** "Field is perpendicular to the **coil plane**"...
    - Then $\vec{B}$ is parallel to $\vec{n}$.
    - $\theta = 0^\circ$ and $\cos(0) = 1$ (Max Flux).
- **If the problem says:** "Field makes angle 30° with the **coil plane**"...
    - Then $\theta = 90^\circ - 30^\circ = 60^\circ$.

Part 3: Electromagnetic Induction

Generating electricity by changing the "Count"

3.1 Lenz's Law (The Direction)

> "The direction of the induced current is such that it opposes the change that produced it."

The Logic Tree for Finding Direction:

1. **Ask:** Is the Flux (Φ) increasing or decreasing?
   - *Example: Magnet moving closer* $\rightarrow \Phi$ *Increases.*
2. **React:**
   - If Φ is **Increasing**: The Loop wants to **fight back**. It creates an induced field ($\vec{B}_{ind}$) in the **OPPOSITE** direction to the external $\vec{B}$.
   - If Φ is **Decreasing**: The Loop wants to **sustain it**. It creates an induced field ($\vec{B}_{ind}$) in the **SAME** direction as the external $\vec{B}$.
3. **Right Hand Rule:** Point your thumb in the direction of $\vec{B}_{ind}$ (from step 2). Your fingers curl in the direction of the induced current (i).

3.2 Faraday's Law (The Magnitude)

The voltage (EMF) generated is the **speed** of flux change.

$$e = -\frac{d\Phi}{dt}$$

- **Average EMF:** $e_{avg} = -\frac{\Delta\Phi}{\Delta t}$
- **Instantaneous EMF:** Derivative (Slope of the $\Phi - t$ graph).

Interpreting the Sign:

- The minus sign mathematically represents Lenz's Law.
- In circuits, we often calculate magnitude $|e| = |\frac{d\Phi}{dt}|$ first, then use Lenz's Law to determine the direction (polarity).

3.3 The "Generator" Equivalent

When a coil has induction, it behaves like a battery with internal resistance (r).

- **Ohm's Law for a Coil:** $u_{AB} = ri - e$
- *Note: This treats the coil as a receiver converting electrical energy to mechanical/heat, or a generator if driven.*

Tech Insight: Why do we care? 🌞

(Context for Solar/Systems)

In **Solar Inverters**, we take DC (constant current) and turn it into AC (alternating current) for your home. How? We rapidly switch the current on and off through a transformer coil.

1. Current changes (di/dt).
2. Magnetic Field changes (dB/dt).
3. Flux changes ($d\Phi/dt$).
4. **Induction happens:** A huge voltage is created in the secondary coil to push power to the grid. *Without induction, there is no grid, no transformers, and no wireless charging.*